

1. *Instructions to the Students:*

2. *All the questions are compulsory.*

3. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*

4. *Use of non-programmable scientific calculators is allowed.*

Assume suitable data wherever necessary and mention it clearly.

(Level CO)

Marks

Q. 1 Solve Any Two of the following.

- A) Write a code to find the largest of three numbers. 6
- B) What is the difference between for loop and while loop?
Write the syntax of for and while loop. 6
- C) Write a code which iterates from numbers 1 to 100, displays "fizz" if the number is divisible by a but not b, displays "buzz" if the number is divisible by b but not a, and displays "fizzbuzz" if the number is divisible by both a and b. a and b are inputs taken from the user. 6

Q.2 Solve Any Two of the following.

- A) Explain crowd computing with an example. 6
- B) Define the list and, using appropriate examples, perform any four operations on the list. 6
- C) Write code to determine whether the input year is a leap year or not. 6

Q. 3 Solve Any Two of the following.

- A) What is the magic square problem?
Write an algorithm for the magic square of size 3. 6
- B) Write a program to define a function named printDict() that can print a dictionary where the keys are numbers between 1 and n (both included) and the values are squares of the keys. The function printDict() doesn't take any arguments. (where n is an integer number). 6
- C) What is bubble sorting? Using the bubble sort algorithm, sort the list [5,1,4,2,8]. 6

Q.4 Solve Any Two of the following.

- A) What is recursion ? Write code for factorial of input number. 6
- B) What is difference between binary search and linear search? 6
- C) What is the dictionary in Python? Describe the functions of a dictionary. 1)get() 2) keys() 3)values() 4)items() 6

Q. 5 Solve Any Two of the following.

- A) Write a program that accepts a sequence of lines as input and prints the lines after making all characters in the sentence capitalized. 6
- B) What is the difference between list and tuples in Python? 6
- C) Write a program to generate a matrix having R rows and C columns such that all the numbers are in increasing order starting from 1 in a row-wise manner. (where R = 3 and C = 3, print the output). 6

*** End ***